INTEGRAL BALANCES & REYNOLDS TRANSPORT

Topic 3: Conservation Laws (Integral Form)

Practical Subchapters & Problem-Solving Guide for 2nd Year Aerospace Engineering

Modular Practical Subchapters

Select a subchapter for in-depth theory, step-by-step algorithms, and canonical exam walkthroughs

SUBCHAPTER 3.1

Kinematic Tool

Reynolds Transport Theorem & Control Volumes

Control volume selection strategy (fixed, moving, deformable), the moving reference frame trick for unsteady waves, and boundary decomposition.

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt}\int_{V_c} \rho b \, dV + \int_{\Sigma_c} \rho b [(\vec{v} - \vec{v}_c)\cdot \vec{n}] \, d\sigma$$

Canonical: The Hydraulic Bore / Moving Shock (CL2)

SUBCHAPTER 3.2

Continuity

Conservation of Mass in Problem Solving

Integral continuity balances, incompressible mass flow conservation, 1D streamtube approximations, and unsteady storage in draining tanks.

$$\frac{d}{dt}\int_{V_c} \rho \, dV + \int_{\Sigma_c} \rho (\vec{v}\cdot\vec{n}) \, d\sigma = 0$$

Canonical: Airfoil Wake Deficit Mass Balance (CL1)

SUBCHAPTER 3.3

Reaction Forces

Linear Momentum & Reaction Forces

Breakdown of the 5 momentum terms, the atmospheric gauge pressure trick, fluid vs solid force conventions, and the 5-stage exam algorithm.

$$\sum \vec{F}_{\text{ext}} = \frac{d}{dt}\int_{V_c} \rho\vec{v}\, dV + \int_{\Sigma_c} \rho\vec{v}[(\vec{v}-\vec{v}_c)\cdot\vec{n}]\, d\sigma$$

Canonical: Pipe Elbow (CL3), Rocket Thrust (CL5), Airfoil Drag (CL1)

SUBCHAPTER 3.4

Turbomachinery

Angular Momentum & Turbomachinery Torque

Strategic pivot selection to eliminate unknown reactions, moment balance of 1D streamtubes, and Euler's turbomachinery equation for rotors.

$$\sum \vec{M}_{\vec{x}_0} = \sum \dot{m}_{\text{out}}(\vec{r}_{\text{out}} \times \vec{v}_{\text{out}}) - \sum \dot{m}_{\text{in}}(\vec{r}_{\text{in}} \times \vec{v}_{\text{in}})$$

Canonical: Pipe Bend Torque (CL3/CL15) & Euler Rotor

SUBCHAPTER 3.5

1st Law

Total Energy Conservation & Thermodynamics

Enthalpy grouping $h = e + p/\rho$, stagnation enthalpy h_0, engineering energy balance with heat transfer, and mechanical shaft work.

$$\dot{Q} + \dot{W}_{\text{shaft}} = \sum \dot{m}_{\text{out}} h_{0,\text{out}} - \sum \dot{m}_{\text{in}} h_{0,\text{in}}$$

Canonical: Mixing Stream Temperature (CL15) & Fan Power (CL10)

SUBCHAPTER 3.6

Navier-Poisson

Stress Tensor, Navier-Poisson & Skin Friction

Cauchy stress decomposition into hydrostatic pressure and viscous deviator $\bar{\bar{\tau}}'$, wall shear stress τ_w, and skin friction drag integration.

$$\bar{\bar{\tau}} = -p\bar{\bar{I}} + 2\mu\bar{\bar{T}}_d + \left(\mu_B - \frac{2}{3}\mu\right)(\nabla \cdot \vec{v})\bar{\bar{I}}$$

Canonical: Wall Friction Drag & Rotating Viscometer Torque

Master Formula Cheat Sheet (Eqs. 3.1 to 3.46)

Complete reference handbook from official UC3M lecture notes

Eq. #	Mathematical Formulation	Physical Meaning & Nomenclature	Hypothesis / Scope
3.1	$\Phi(t) = \int_{V_f(t)} \phi(\vec{x}, t)\, dV$	Definition of extensive property from volumetric density $\phi = \rho, \rho\vec{v}, \rho(e + v^2/2)$.	Closed material system $V_f(t)$.
3.2	$\frac{d}{dt}\left[\int_{V_f(t)} \phi dV\right] = \lim_{\Delta t \to 0} \frac{1}{\Delta t}\left[\int_{V_f(t+\Delta t)} \phi(t+\Delta t) dV - \int_{V_f(t)} \phi(t) dV\right]$	Formal limit definition of ordinary material derivative of volume integral.	General continuum mechanics.
3.3	$\lim_{\Delta t \to 0} \frac{1}{\Delta t}\left[\int_{V_f(t)} [\phi(t+\Delta t) - \phi(t)] dV + \int_{V_f(t+\Delta t) - V_f(t)} \phi(t+\Delta t) dV\right]$	Decomposition into internal unsteadiness storage and boundary swept volume.	Domain splitting.

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3.4	$$\lim_{\Delta t \to 0} \frac{1}{\Delta t} \int_{V_f(t)} [\phi(t+\Delta t) - \phi(t)] dV = \int_{V_f(t)} \frac{\partial \phi}{\partial t} dV$$	Internal limit contribution: local unsteadiness within instantaneous volume.	1st-order Taylor expansion.
3.5	$$\lim_{\Delta t \to 0} \frac{1}{\Delta t} \int_{V_f(t+\Delta t) - V_f(t)} \phi(t+\Delta t) dV = \int_{\Sigma_f(t)} \phi \vec{v} \cdot \vec{n} \, d\sigma$$	Swept volume limit: net convective flux escaping through moving boundary $\Sigma_f(t)$.	Differential geometry: $dV = (\vec{v}\Delta t) \cdot \vec{n} d\sigma$.
3.6	$$\frac{d}{dt} \left[\int_{V_f(t)} \phi \, dV \right] = \int_{V_f(t)} \frac{\partial \phi}{\partial t} \, dV + \int_{\Sigma_f(t)} \phi \, \vec{v} \cdot \vec{n} \, d\sigma$$	**Reynolds Transport Theorem (RTT)** for a closed material fluid volume $V_f(t)$.	Material boundary moving at fluid velocity $\vec{v}$.
3.7	$$\frac{d}{dt} \left[\int_{V_c(t)} \phi \, dV \right] = \int_{V_c(t)} \frac{\partial \phi}{\partial t} \, dV + \int_{\Sigma_c(t)} \phi \, \vec{v}_c \cdot \vec{n} \, d\sigma$$	RTT applied to an arbitrary moving geometric control volume $V_c(t)$ with surface speed $\vec{v}_c$.	Arbitrary geometric volume.
3.8	$$\frac{d}{dt} \left[\int_{V_f(t)} \phi \, dV \right] = \frac{d}{dt} \left[\int_{V_c(t)} \phi \, dV \right] + \int_{\Sigma_c(t)} \phi \, (\vec{v} - \vec{v}_c) \cdot \vec{n} \, d\sigma$$	**Universal Reynolds Relation:** connects material derivative to moving control volume.	Relative convective velocity $(\vec{v} - \vec{v}_c)$.
3.9	$$\frac{d}{dt} \left[\int_{V_f(t)} \rho \, dV \right] = 0$$	Principle of mass conservation for a closed material fluid volume.	Classical non-relativistic mechanics.
3.10	$$\frac{d}{dt} \left[\int_{V_c(t)} \rho \, dV \right] + \int_{\Sigma_c(t)} \rho (\vec{v} - \vec{v}_c) \cdot \vec{n} \, d\sigma = 0$$	**Integral Continuity Equation** for an arbitrary moving control volume $V_c(t)$.	Moving control volume.

Eq. #	Mathematical Formulation	Physical Meaning & Nomenclature	Hypothesis / Scope
3.11	$$\int_{V_0} \frac{\partial \rho}{\partial t}\, dV + \int_{\Sigma_0} \rho \vec{v} \cdot \vec{n}\, d\sigma = 0$$	Mass conservation for a fixed spatial control volume ($V_0, \vec{v}_c = 0$).	Rigid, stationary control volume.
3.12	$$d\vec{F}_m = \rho \vec{f}_m(\vec{x}, t)\, dV$$	Elemental volumetric (body / mass) force acting on fluid particle.	Long-range external forces.
3.13	$$\vec{f}_m = \vec{g} - \vec{a}_0 - \frac{d\vec{\Omega}}{dt} \wedge \vec{x} - \vec{\Omega} \wedge (\vec{\Omega} \wedge \vec{x}) - 2\vec{\Omega} \wedge \vec{v}$$	Body force in non-inertial frame: gravity + translation + angular + centrifugal + Coriolis.	Accelerating / rotating frame.
3.14	$$\vec{g} - \vec{a}_0 - \vec{\Omega} \wedge (\vec{\Omega} \wedge \vec{x}) = -\nabla U$$	Conservative potential $U = -\vec{g} \cdot \vec{x} + \vec{a}_0 \cdot \vec{x} - \frac{1}{2}\|\vec{\Omega} \wedge \vec{x}\|^2$.	Steady rotation and linear translation.
3.15	$$d\vec{F}_s = \vec{f}_n(\vec{n}, \vec{x}, t)\, d\sigma$$	Surface contact force (stress vector $\vec{f}_n$) across surface element with normal $\vec{n}$.	Intermolecular contact force.
3.16	$$dA\vec{f}_n - dA_1\vec{f}_1 - dA_2\vec{f}_2 - dA_3\vec{f}_3 = 0$$	Force equilibrium on Cauchy's differential tetrahedron as size $h \to 0$.	Surface forces $\sim h^2 \gg$ mass forces $\sim h^3$.
3.17	$$\vec{f}_n = \bar{\bar{\tau}} \cdot \vec{n}$$	**Cauchy's Stress Principle:** linear mapping from normal $\vec{n}$ to stress vector $\vec{f}_n$.	Classical continuum media.
3.18	$$\bar{\bar{\tau}} = [\tau_{ij}]$$	Components of Cauchy stress tensor in Cartesian coordinate system.	3x3 tensor matrix.

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3.19	$\tau_{ij} = \tau_{ji}$	**Symmetry of Cauchy stress tensor:** proved via angular momentum balance on cube.	No body couples / micropolar effects.
3.20	$\bar{\bar{\tau}} \cdot \vec{n} = \lambda \vec{n}$	Eigenvalue equation for principal stress directions and magnitudes (zero shear stress).	Principal axes of stress.
3.21	$\lvert \bar{\bar{\tau}} - \lambda \bar{\bar{I}} \rvert = 0$	Secular characteristic equation: yields 3 real eigenvalues and orthogonal eigenvectors.	Guaranteed by symmetry of $\bar{\bar{\tau}}$.
3.22	$\vec{F}_s = \int_\Sigma \bar{\bar{\tau}} \cdot \vec{n} \, d\sigma$	Resultant surface force acting on a finite surface Σ.	Any fluid surface.
3.23	$\int_\Sigma \bar{\bar{\tau}} \cdot \vec{n} \, d\sigma = \int_V (\nabla \cdot \bar{\bar{\tau}}) \, dV$	Divergence theorem: $\nabla \cdot \bar{\bar{\tau}}$ represents net surface force per unit volume.	Closed boundary enclosing volume V.
3.24	$\rho dV \frac{D\vec{v}}{Dt} = (\nabla \cdot \bar{\bar{\tau}}) dV + \rho \vec{f}_m dV$	Newton's second law applied to an infinitesimal fluid particle.	Fluid parcel element.
3.25	$\rho \frac{D\vec{v}}{Dt} = \nabla \cdot \bar{\bar{\tau}} + \rho \vec{f}_m$	**Cauchy's Differential Equation of Motion** for any continuum medium.	Universal continuum balance.
3.26	$\vec{f}_n = -p\vec{n}$	Purely normal hydrostatic stress in fluid at rest or in rigid body motion.	Zero velocity gradients ($\bar{\bar{T}}_d = 0$).

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3.27	$\bar{\bar{\tau}} = -p\bar{\bar{I}}$	Isotropic spherical hydrostatic stress tensor.	Static fluid state.
3.28	$\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$	Stress decomposition into thermodynamic pressure p and viscous deviatoric stress $\bar{\bar{\tau}}'$.	Moving fluid.
3.29	$\tau'_{ij} = \alpha_{ijkl}\gamma_{kl}$	Newtonian fluid hypothesis: linear constitutive relationship with strain rate tensor.	Newtonian fluid assumption.
3.30	$\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d + \lambda(\nabla\cdot\vec{v})\bar{\bar{I}}$	Navier-Poisson equation with dynamic shear viscosity μ and second viscosity λ.	Isotropic Newtonian fluid.
3.31	$\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d + \left(\mu_B - \frac{2}{3}\mu\right)(\nabla\cdot\vec{v})\bar{\bar{I}}$	**Canonical Navier-Poisson Equation:** formulated with bulk viscosity $\mu_B = \lambda + \frac{2}{3}\mu$.	Stokes hypothesis: $\mu_B \approx 0 \implies \lambda = -\frac{2}{3}\mu$.
3.32	$\vec{F} = -\int_\Sigma p\vec{n}d\sigma + \int_\Sigma \bar{\bar{\tau}}'\cdot\vec{n}d\sigma$	**Aerodynamic Force on Submerged Body:** pressure lift/form drag + viscous skin friction.	Wetted body surface ($\vec{n}$ outwards into fluid).
3.33	$\vec{M}_{\vec{x}_0} = -\int_\Sigma (\vec{x}-\vec{x}_0)\wedge(p\vec{n})d\sigma + \int_\Sigma (\vec{x}-\vec{x}_0)\wedge(\bar{\bar{\tau}}'\cdot\vec{n})d\sigma$	**Aerodynamic Moment on Submerged Body** about reduction point $\vec{x}_0$.	Pitching / yawing moments.
4.2	$\frac{d}{dt}\left[\int_{V_f}\rho\vec{v}dV\right] = -\int_{\Sigma_f}p\vec{n}d\sigma + \int_{\Sigma_f}\bar{\bar{\tau}}'\cdot\vec{n}d\sigma + \int_{V_f}\rho\vec{f}_m dV$	Newton's 2nd law for a closed material fluid volume $V_f(t)$ (Eq. 3.34 with Eq. 3.28 inserted; Eq. 3.34 itself uses $\bar{\bar{\tau}}\cdot\vec{n}$).	Closed fluid system.

Eq. #	Mathematical Formulation	Physical Meaning & Nomenclature	Hypothesis / Scope
3.35	$\frac{d}{dt}\left[\int_{V_c}\rho\vec{v}dV\right]+\int_{\Sigma_c}\rho\vec{v}[(\vec{v}-\vec{v}_c)\cdot\vec{n}]d\sigma=-\int_{\Sigma_c}p\vec{n}d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\vec{n}d\sigma+\int_{V_c}\rho\vec{f}_m dV$	**Integral Linear Momentum Conservation** for arbitrary moving control volume $V_c(t)$.	Fundamental equation for thrust & forces.
3.36	$\int_{V_0}\frac{\partial(\rho\vec{v})}{\partial t}dV+\int_{\Sigma_0}\rho\vec{v}(\vec{v}\cdot\vec{n})d\sigma=-\int_{\Sigma_0}p\vec{n}d\sigma+\int_{\Sigma_0}\bar{\bar{\tau}}'\cdot\vec{n}d\sigma+\int_{V_0}\rho\vec{f}_m dV$	Linear momentum conservation for fixed spatial control volume ($V_0, \vec{v}_c=0$).	Pipes, nozzles, wind tunnels.
3.37	$\frac{d}{dt}\left[\int_{V_f}\rho(\vec{x}-\vec{x}_0)\wedge\vec{v}dV\right]=\sum\vec{M}_{\vec{x}_0,\text{ext}}$	Angular momentum conservation for material volume $V_f(t)$.	Closed material mass.
3.38	$\frac{d}{dt}\left[\int_{V_c}\rho[(\vec{x}-\vec{x}_0)\wedge\vec{v}]dV\right]+\int_{\Sigma_c}\rho[(\vec{x}-\vec{x}_0)\wedge\vec{v}][(\vec{v}-\vec{v}_c)\cdot\vec{n}]d\sigma=\sum\vec{M}_{\vec{x}_0,\text{ext}}$	**Integral Angular Momentum Conservation** for moving control volume $V_c(t)$.	Turbomachinery rotors, fans, elbows.
3.39	$d\dot{Q}_{\text{cond}}=q_n(\vec{n},\vec{x},t)d\sigma$	Heat conduction rate per unit area across surface element in direction $\vec{n}$.	Molecular thermal transfer.
3.40	$q_n dA=q_1 dA_1+q_2 dA_2+q_3 dA_3$	Thermal equilibrium in Cauchy's tetrahedron as size $h\to 0$.	Local heat flux equilibrium.
3.41	$q_n=\vec{q}\cdot\vec{n}$	Thermal Cauchy relation: scalar heat flux is projection of heat flux vector $\vec{q}$.	Continuous thermal medium.
3.42	$\dot{Q}_{\text{cond}}=\int_{\Sigma}\vec{q}\cdot\vec{n}d\sigma$	Total heat conducted through a finite surface Σ.	Thermal boundaries.

Eq. #	Mathematical Formulation	Physical Meaning & Nomenclature	Hypothesis / Scope
3.43	$$\int_{\Sigma} \vec{q} \cdot \vec{n} d\sigma = \int_{V} (\nabla \cdot \vec{q}) dV$$	Thermal Gauss theorem: $\nabla \cdot \vec{q}$ is net heat loss rate per unit volume.	Closed boundary.
3.44	$$\vec{q} = -k \nabla T$$	**Fourier's Law of Heat Conduction:** heat flows against the thermal gradient.	Isotropic thermal conductivity k.
3.45	$$\frac{d}{dt}\left[\int_{V_f} \rho(e + \frac{\|\vec{v}\|^2}{2}) dV\right] = -\int_{\Sigma_f} p\vec{v} \cdot \vec{n} d\sigma + \int_{\Sigma_f} \vec{v} \cdot \vec{\bar{\tau}}' \cdot \vec{n} d\sigma + \int_{V_f} \rho \vec{f}_m \cdot \vec{v} dV - \int_{\Sigma} \vec{q} \cdot \vec{n} d\sigma + \int_{V_f} (Q_c + Q_r) dV$$	**First Law of Thermodynamics** for closed material fluid volume $V_f(t)$.	Conservation of total energy.
3.46	$$\frac{d}{dt}\left[\int_{V_c} \rho(e + \frac{\|\vec{v}\|^2}{2}) dV\right] + \int_{\Sigma_c} \rho(e + \frac{\|\vec{v}\|^2}{2})[(\vec{v} - \vec{v}_c) \cdot \vec{n}] d\sigma = -\int_{\Sigma_c} p\vec{v} \cdot \vec{n} d\sigma + \int_{\Sigma_c} \vec{v} \cdot \vec{\bar{\tau}} \cdot \vec{n} d\sigma + \int_{V_c} \rho \vec{f}_m \cdot \vec{v} dV - \int_{\Sigma_c} \vec{q} \cdot \vec{n} d\sigma + \int_{V_c} (Q_c + Q_r) dV$$	**Integral Total Energy Conservation** for arbitrary moving control volume $V_c(t)$.	Master energy balance with work & heat.